

DESIGN
PROJECTS

MECHANICAL
ENGINEERING

PORTFOLIO

2023
2027

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TITLE: Evaluating Whole-Body Vibration Effects on the Lumbar Spine Using Finite Element Modeling

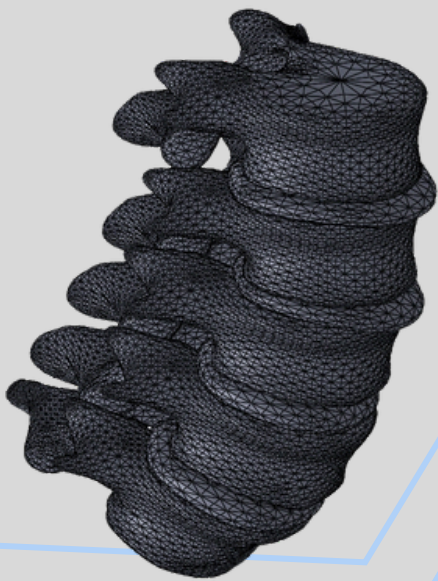


FIGURE 1. Lumbar Spine Model separated from the human spinal model

Problem:

Whole-body vibration (WBV), commonly experienced in vehicles and occupational environments, is a significant contributor to lower back pain and lumbar spine disorders. Studying its mechanical effects directly on the human spine is difficult due to ethical, safety, and experimental limitations. As a result, understanding how vibration-induced loads affect spinal stress and deformation remains challenging.

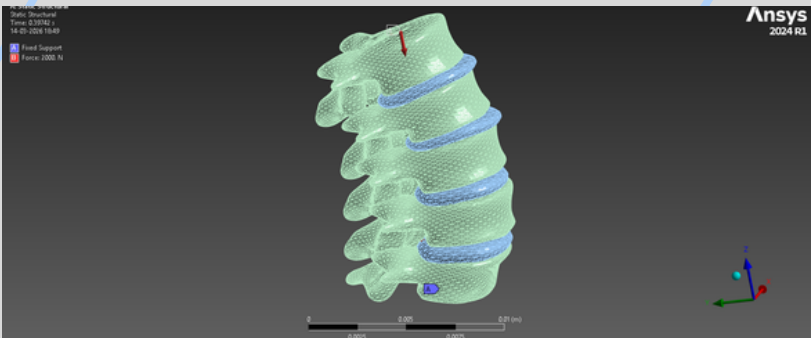


FIGURE 2. Materials assigned to the parts of the lumbar spine model

Project Workflow

- Lumbar spine model obtained from anatomical 3D data
- Geometry cleaned and simplified in Blender
- Converted to solid body in SolidWorks
- Imported into ANSYS Workbench
- Mesh generated and material properties assigned
- Static structural simulations performed

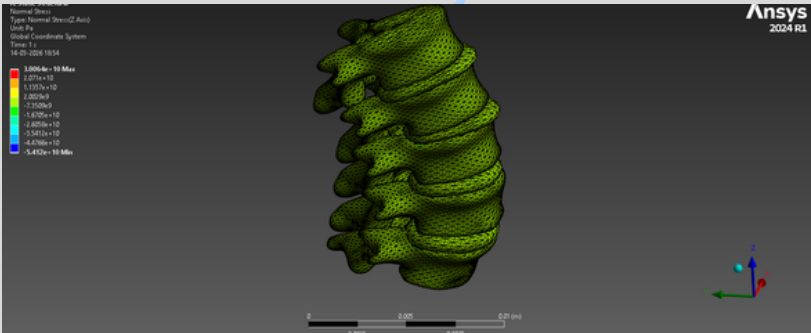


FIGURE 3. Normal stress when variable forces were applied.

Simulation Setup

The simplified lumbar spine model (**L1-L5**) was meshed with a **0.7 mm** element size, resulting in approximately **188k nodes** and **106k elements**. The **L5 vertebra** was fixed to represent pelvic support, while **axial loads** of **300 N**, **1000 N**, and **3000 N** were applied at **L1** to simulate physiological compressive loading. Simulations were performed in **ANSYS Workbench** to analyze stress distribution and deformation.

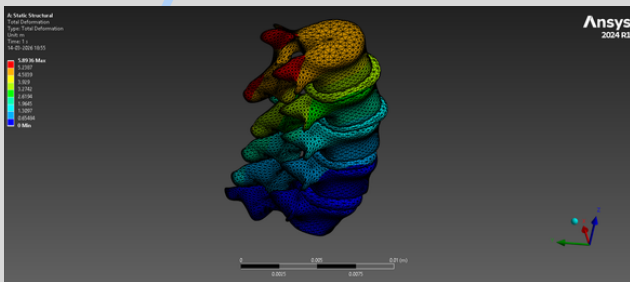


FIGURE 4. Total deformation when variable forces were applied.

Solution:

To address this, a simplified finite element model of the lumbar spine (L1-L5) was developed and analyzed using ANSYS Workbench. The simulation evaluated stress distribution and deformation under different compressive loads, helping to understand how vibration-related forces influence spinal biomechanics and potential injury risks.

TITLE: Silico Damp



FIGURE 1. Normal bike centre stand along with dampeners

Problem:

Motorcycle centre and side stands generate impact noise when they deploy or retract due to the sudden contact between the stand stopper and the frame. This impact force leads to vibration, noise generation, and reduced ride comfort. The challenge is to reduce this noise without significantly modifying the existing stand design or increasing manufacturing cost.

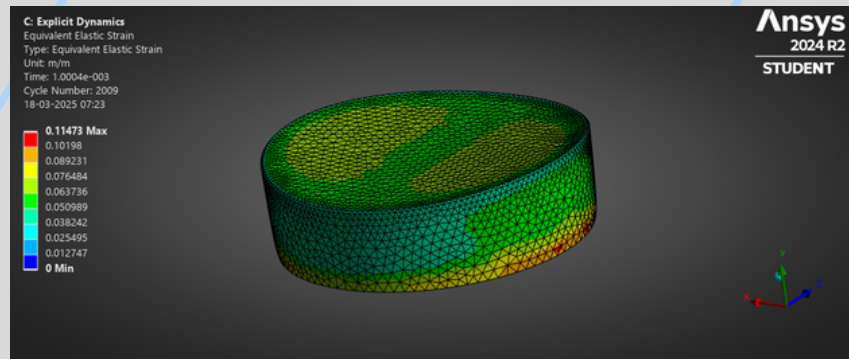


FIGURE 2. Impact during closure (High dampening Silicone rubber pad)

Project Workflow

- Identified impact noise issue in motorcycle centre stand operation.
- Developed CAD model and proposed a high-damping silicone pad at the stopper contact point.
- Performed CAE simulations to analyze impact force and energy absorption.
- Compared results with standard rubber dampers and evaluated manufacturing feasibility.

Material Comparison

Compared to the standard EPDM rubber pad (**60–70A hardness**), the proposed high-damping silicone rubber pad (**40–50A hardness**) provides improved damping performance. While the conventional rubber operates between **–10°C to 70°C** and offers **10–15%** noise reduction, the silicone pad functions across a wider **–40°C to 200°C** range and achieves approximately **30–40%** noise reduction due to higher strain energy absorption. Additionally, the silicone material is fully waterproof, non-porous, and easier to clean, and is implemented with a slightly larger **8–12 mm** thickness compared to the **5–7 mm** standard rubber pad for improved impact damping.

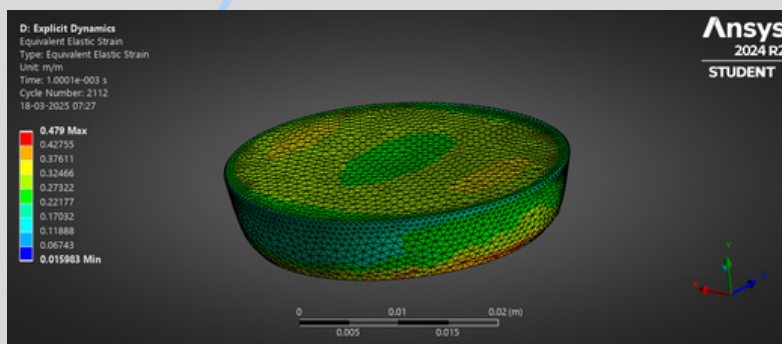


FIGURE 3. Impact during opening (High dampening Silicone rubber pad)

Solution:

A high-damping silicone elastomer pad was introduced at the stopper contact point to absorb impact energy during stand motion. CAD modelling and CAE simulations showed that the silicone pad undergoes greater strain, converting more impact energy into strain energy, which reduces transmitted force and minimizes noise during stand deployment and closure.

TITLE: MotoPulse



FIGURE 1. Normal cycle model fitted with the electric conversion kit

Problem:

The project focuses on developing an electric conversion kit for bicycles that requires minimal modifications and can be universally installed on different cycles, while also addressing the issue of unwanted pedal rotation that occurs when using a mid-drive motor system.

Technical Specifications

- Motor Power: **250 W** mid-drive electric motor.
- Battery: **24 V, 7.5 Ah** lithium battery pack.
- Controller: **15 A** motor controller with pedal assist system (PAS).
- Estimated Range: Approximately **15–18 km** per charge under ideal conditions.
- Charging Time: Approximately **3.5–4** hours using a standard charger.

Key Features

- Universal electric conversion kit compatible with most standard bicycles.
- Mid-drive motor configuration providing higher torque and improved drivetrain efficiency.
- Custom pedal shaft assembly with an integrated mechanical pin-lock mechanism.
- Improved hill-climbing capability and better performance on varied terrains.
- Compact and lightweight drivetrain integration with minimal cycle modification.

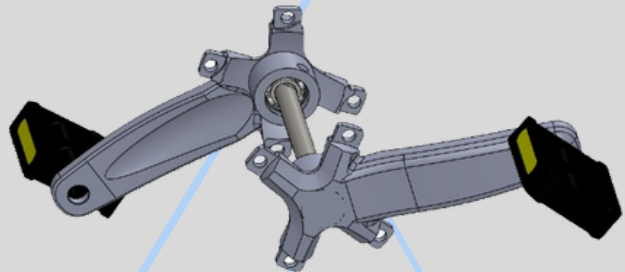


FIGURE 2. Custom spindle and pedal shaft assembly

Solution:

A mid-drive electric conversion kit is proposed because it integrates with the bicycle's existing drivetrain, providing higher efficiency, better torque for hill climbing, and improved performance on varied terrains; additionally, a mechanical pin-lock mechanism is incorporated in the pedal shaft assembly to lock or release pedal rotation, allowing the rider to stop pedal movement during motor operation and re-engage it when pedaling is required.